

Agile Plan — Kids Clinic Management System

Scrum Methodology · IEEE 830-1998 Compliant · v2.0

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1. Scrum Framework

The original Waterfall methodology has been replaced with **Agile Scrum** to enable iterative development, continuous stakeholder feedback, and adaptive planning.

1.1 Why Agile / Scrum

Aspect	Scrum Implementation
Iterations	2-week Sprints delivering potentially shippable increments
Requirements	Living Product Backlog refined through ongoing stakeholder collaboration
Testing	Continuous integration testing within each Sprint
Delivery	Incremental feature releases enabling early value realization
Flexibility	Sprint Retrospectives drive process improvement and adaptation
Stakeholder Engagement	Sprint Reviews provide regular demo and feedback opportunities

1.2 Scrum Roles

Role	Who	Responsibilities
Product Owner	The Doctor	Defines priorities, accepts/rejects deliverables, manages Product Backlog
Scrum Master	Project Team Lead	Facilitates ceremonies, removes impediments, coaches the team
Development Team	Full-stack developers + QA	Self-organizing, cross-functional, delivers working increments

1.3 Scrum Ceremonies

Ceremony	Duration	Purpose
Sprint Planning	4 hours	Select and commit to Sprint Backlog items from Product Backlog
Daily Stand-up	15 min	Synchronize activities, identify blockers, adjust plan
Sprint Review	2 hours	Demo increment to stakeholders, gather feedback
Sprint Retrospective	1.5 hours	Inspect and adapt processes for next Sprint
Backlog Refinement	2 hours/week	Estimate, prioritize, and clarify upcoming items

2. User Stories & Acceptance Criteria

Each story follows the **Connextra format**: *"As a [role], I want [feature], so that [benefit]."* Acceptance Criteria use **Given-When-Then** format. Story Points follow the **Fibonacci scale**: 1 · 2 · 3 · 5 · 8.

 Sprint 1 — Foundation: Authentication & Core Setup

US-001 · Login to the System

3 SP · High · REQ-001

As a Doctor or Employee, I want to log in with my username and password, so that I can securely access the system.

#	Acceptance Criterion
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- | | |
|-------|---|
| AC1.1 | Given the user is on the login page, When they enter valid credentials and click Login, Then they are redirected to the dashboard within 3 seconds. |
| AC1.2 | Given the user enters invalid credentials, When they click Login, Then an error message <i>"Invalid username or password"</i> is displayed without revealing which field is incorrect. |
| AC1.3 | Given the user is authenticated, When 30 minutes of inactivity passes, Then the session expires and the user is redirected to the login page. |

US-002 · Manage User Accounts

| 5 SP · High · REQ-032

As a Doctor, I want to create and manage user accounts, **so that** I can control system access for clinic staff.

#	Acceptance Criterion
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|-------|--|
| AC2.1 | Given the Doctor is logged in, When they navigate to User Management and fill in username, password, and role, Then a new user is created and can immediately log in. |
| AC2.2 | Given the Doctor views the user list, When they click Edit on a user, Then they can modify the user's role and save changes. |
| AC2.3 | Given the Doctor attempts to delete a user, When the user has associated records, Then deletion is prevented with a message explaining the dependency. |

🧑 Sprint 2 — Patient Management Core

US-003 · Add a New Child's Profile

| 5 SP · High · REQ-006

As a Doctor or Employee, I want to add a new child's medical profile, **so that** the clinic has complete patient records.

#	Acceptance Criterion
AC3.1	Given the user is logged in, When they navigate to Add Child and enter required fields (name, DOB, gender, parent name, phone), Then the record is saved and a success message is displayed.
AC3.2	Given required fields are missing, When the user submits the form, Then validation errors appear next to each missing field.
AC3.3	Given a child with the same name and DOB exists, When the user attempts to add, Then a duplicate warning is shown with a link to the existing record.

US-004 • View List of Registered Children

3 SP • High • REQ-005

As a Doctor or Employee, I want to view a list of all registered children, **so that** I can quickly find patient records.

#	Acceptance Criterion
AC4.1	Given the user is logged in, When they click "Children's List", Then a paginated table shows all children sorted by registration date (newest first).
AC4.2	Given the list has more than 20 records, When displayed, Then pagination controls are available.
AC4.3	Given the list is empty, When displayed, Then an informational message appears with a link to add the first child.

US-005 • Search for Children by Name

2 SP • High • REQ-009

As a Doctor or Employee, I want to search for children by name, **so that** I can quickly locate a specific patient.

#	Acceptance Criterion
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AC5.1	Given the user is on the Children's List page, When they type 3+ characters in the search box, Then results filter in real-time showing matching names.
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AC5.2	Given no results match the search term, When the user searches, Then a "No results found" message is displayed with an option to clear the search.
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US-006 • Edit or Delete a Child's Record

3 SP • High • REQ-007, REQ-008

As a Doctor or Employee, **I want to** edit or delete a child's record, **so that** I can correct errors or remove obsolete entries.

#	Acceptance Criterion
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AC6.1	Given the user selects a child record, When they click Edit, Then a pre-filled form appears allowing modification of all fields.
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AC6.2	Given the user clicks Delete, When a confirmation dialog appears and they confirm, Then the record is soft-deleted (hidden but recoverable).
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AC6.3	Given the child has linked prescriptions, When the user attempts deletion, Then deletion is blocked with a message listing the dependencies.
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Sprint 3 — Visit Scheduling

US-007 • Schedule a Medical Visit

5 SP • High • REQ-010

As a Doctor or Employee, **I want to** schedule a medical visit for a child, **so that** appointments are properly organized.

#	Acceptance Criterion
AC7.1	Given a child record exists, When the user clicks "Schedule Visit" and enters date, time, type, and notes, Then the visit is saved and appears in the child's history.
AC7.2	Given the selected time slot conflicts with an existing visit, When the user submits, Then a conflict warning is displayed with overlapping visit details.
AC7.3	Given the visit date is in the past, When the user submits, Then a validation error prevents scheduling with a clear message.

US-008 • View Today's Scheduled Visits

3 SP • High • REQ-011

As a Doctor or Employee, I want to view today's scheduled visits, **so that** I can prepare for the day's appointments.

#	Acceptance Criterion
AC8.1	Given the user is logged in, When they open the dashboard or "Today's Visits", Then all visits for the current date are displayed sorted by time.
AC8.2	Given there are no visits today, When the page loads, Then an empty state message appears with a button to schedule a visit.

US-009 • Edit or Cancel a Scheduled Visit

5 SP • High • REQ-013, REQ-014

As a Doctor or Employee, I want to edit or cancel a scheduled visit, **so that** I can accommodate changes in the schedule.

#	Acceptance Criterion
AC9.1	Given a visit exists, When the user clicks Edit, Then all visit details are editable and changes are saved with an audit trail.
AC9.2	Given a visit exists, When the user clicks Cancel/Delete and confirms, Then the visit is removed from the schedule and the child's history notes the cancellation.

US-010 • View a Child's Complete Visit History

3 SP • Medium • REQ-012

As a Doctor or Employee, I want to view a child's complete visit history, so that I can review past medical encounters.

#	Acceptance Criterion
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AC10.1	Given a child record is selected, When the user clicks "Visit History", Then all past visits are displayed in reverse chronological order.
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AC10.2	Given the history is long, When displayed, Then results are paginated with filters for date range and visit type.
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Sprint 4 — Consultation Management

US-011 • Schedule a Consultation Linked to a Visit

5 SP • High • REQ-015

As a Doctor or Employee, I want to schedule a consultation linked to a visit, so that medical encounters are properly documented.

#	Acceptance Criterion
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AC11.1	Given a visit is selected, When the user clicks "Add Consultation" and enters symptoms, diagnosis, notes, and doctor, Then the consultation is saved and linked to the visit.
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AC11.2	Given a consultation is added, When the visit list is viewed, Then a visual indicator shows that the visit has an associated consultation.
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US-012 • View Consultation History

3 SP • Medium • REQ-016

As a Doctor or Employee, I want to view consultation history, **so that** I can review past diagnoses and treatments.

#	Acceptance Criterion
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AC12.1	Given a child or visit is selected, When the user clicks "Consultation History", Then all consultations are displayed with doctor, date, diagnosis, and symptoms.
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AC12.2	Given consultations exist, When viewing history, Then each entry is expandable to show full details including attached prescriptions.
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Sprint 5 — Prescription Management (Doctor Only)

US-013 · Create a Prescription

┆ 5 SP · High · REQ-017, REQ-022

As a Doctor, I want to create a prescription for a consultation, **so that** medications are properly documented and printable.

#	Acceptance Criterion
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AC13.1	Given a consultation is selected, When the Doctor clicks "Add Prescription" and enters medication, dosage, frequency, duration, and instructions, Then the prescription is saved and linked to the consultation.
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AC13.2	Given an Employee attempts to add a prescription, When they access the prescription form, Then access is denied with a role-appropriate message.
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AC13.3	Given required prescription fields are missing, When the Doctor submits, Then validation errors appear for each missing field.
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US-014 · Edit or Delete a Prescription

┆ 3 SP · Medium · REQ-018, REQ-019

As a Doctor, I want to edit or delete a prescription, **so that** I can correct errors or cancel prescriptions.

#	Acceptance Criterion
AC14.1	Given a prescription exists and is not dispensed, When the Doctor clicks Edit, Then the prescription is editable and changes are saved.
AC14.2	Given a prescription exists, When the Doctor clicks Delete and confirms, Then the prescription is removed with an audit log entry.

US-015 • Print a Prescription

| 3 SP • High • REQ-020

As a Doctor, I want to print a prescription, so that I can provide a physical copy to the patient's guardian.

#	Acceptance Criterion
AC15.1	Given a prescription exists, When the Doctor clicks Print, Then a print-ready page opens with clinic letterhead, patient details, and prescription formatted for standard A4 paper.
AC15.2	Given the print dialog, When the Doctor confirms, Then the prescription prints correctly with all fields legible.

US-016 • View All Prescription Records

| 3 SP • Medium • REQ-021

As a Doctor, I want to view all prescription records, so that I can track medication history for any patient.

#	Acceptance Criterion
AC16.1	Given the Doctor navigates to Prescriptions, When the page loads, Then a filterable list shows all prescriptions with patient name, medication, date, and doctor.
AC16.2	Given the Doctor filters by patient name, When applied, Then only prescriptions for that patient are shown.

US-017 • Manage the Vaccine Catalog

| 5 SP • High • REQ-028

As a Doctor or Employee, I want to manage the vaccine catalog, so that the system maintains an accurate list of available vaccines.

#	Acceptance Criterion
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AC17.1	Given the user is logged in, When they navigate to Vaccine Catalog and add a new vaccine with name, type, and manufacturer, Then the vaccine appears in the catalog.
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AC17.2	Given a vaccine is linked to historical records, When the user attempts to delete it, Then deletion is prevented with a message explaining the dependency.
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US-018 • Record a Vaccination for a Child

| 5 SP • High • REQ-023

As a Doctor or Employee, I want to record a vaccination for a child, so that immunization history is properly tracked.

#	Acceptance Criterion
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AC18.1	Given a child record exists, When the user selects a vaccine from the catalog, enters batch number, date, and next due date, Then the vaccination record is saved.
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AC18.2	Given a vaccination is recorded with a next due date, When that date approaches, Then the child appears in upcoming vaccination reminders.
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US-019 • View a Child's Vaccination History

| 3 SP • High • REQ-024

As a Doctor or Employee, I want to view vaccination history, so that I can verify immunization completeness.

#	Acceptance Criterion
AC19.1	Given a child record is selected, When the user clicks "Vaccination History", Then all vaccinations are displayed in chronological order with vaccine name, date, and next due date.
AC19.2	Given vaccinations exist, When viewing history, Then overdue vaccinations are highlighted in red for immediate attention.

US-020 • View Today's Vaccination Records

3 SP • Medium • REQ-025, REQ-026

As a Doctor or Employee, I want to view today's vaccination records, so that I can prepare and verify daily immunization activities.

#	Acceptance Criterion
AC20.1	Given the current date, When the user opens "Today's Vaccinations", Then all vaccinations scheduled or administered today are displayed.
AC20.2	Given a vaccination record is displayed, When the user clicks Edit, Then the record becomes editable with validation.

Sprint 7 — Financial Management & Reporting

US-021 • View Daily Account Summary

3 SP • Medium • REQ-029

As a Doctor, I want to view daily account summaries, so that I can track daily revenue and visit volume.

#	Acceptance Criterion
AC21.1	Given the Doctor navigates to Daily Accounts, When the page loads, Then a summary shows the selected date, total visits, and total revenue.
AC21.2	Given no visits occurred on the selected date, When the summary is viewed, Then zero values are displayed with an informational message.

US-022 • View Monthly Account Summary

3 SP • Medium • REQ-030

As a Doctor, I want to view monthly account summaries, so that I can analyze monthly performance trends.

#	Acceptance Criterion
AC22.1	Given the Doctor selects a month and year, When they view the summary, Then total visits, total revenue, and average revenue per visit are displayed.
AC22.2	Given the summary is viewed, When data exists, Then a comparison with the previous month is shown as a percentage change.

US-023 • Restrict Financial Access to Doctor Only

3 SP • High • REQ-031

As a Doctor, I want financial summaries to be accessible only to me, so that financial data remains confidential.

#	Acceptance Criterion
AC23.1	Given an Employee attempts to access the Accounts section, When they navigate to it, Then access is denied and they are redirected to the dashboard.
AC23.2	Given the Doctor accesses financial data, When viewing, Then all monetary values are formatted to 2 decimal places with currency symbol.

US-024 · Graceful Error Handling

| 3 SP · Medium · NFR-013

As a user, I want the system to handle errors gracefully, **so that** I can recover from unexpected issues.

#	Acceptance Criterion
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AC24.1	Given any system error occurs, When detected, Then a user-friendly error page is shown with options to retry or return home.
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AC24.2	Given a database connection fails, When the user performs an action, Then the error is logged and a maintenance message is displayed.
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US-025 · Audit Logging for All Data Operations

| 5 SP · Medium · NFR-008

As a user, I want all data operations to be logged, **so that** the clinic maintains an audit trail.

#	Acceptance Criterion
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AC25.1	Given any create, update, or delete operation, When performed, Then an audit log entry records the user, timestamp, action type, and affected record.
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AC25.2	Given the audit log, When viewed by a Doctor, Then entries are filterable by date range, user, and action type.
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3. Product Backlog

| **Total: 92 Story Points** across 25 stories · Ordered by business value and dependency.

Rank	ID	User Story	SP	Priority	Sprint
1	US-001	Login to system	3	High	S1
2	US-002	Manage user accounts	5	High	S1
3	US-004	View children's list	3	High	S2
4	US-003	Add child record	5	High	S2
5	US-005	Search children by name	2	High	S2
6	US-006	Edit / delete child record	3	High	S2
7	US-007	Schedule medical visit	5	High	S3
8	US-008	View today's visits	3	High	S3
9	US-009	Edit / cancel visit	5	High	S3
10	US-010	View visit history	3	Medium	S3
11	US-011	Schedule consultation	5	High	S4
12	US-012	View consultation history	3	Medium	S4
13	US-013	Create prescription	5	High	S5
14	US-014	Edit / delete prescription	3	Medium	S5
15	US-015	Print prescription	3	High	S5
16	US-016	View prescription log	3	Medium	S5
17	US-017	Manage vaccine catalog	5	High	S6
18	US-018	Record vaccination	5	High	S6
19	US-019	View vaccination history	3	High	S6
20	US-020	View today's vaccinations	3	Medium	S6
21	US-021	Daily account summary	3	Medium	S7
22	US-022	Monthly account summary	3	Medium	S7
23	US-023	Financial access control	3	High	S7

Rank	ID	User Story	SP	Priority	Sprint
24	US-024	Error handling & recovery	3	Medium	S8
25	US-025	Audit logging system	5	Medium	S8

4. Sprint Planning

Sprint Length: 2 weeks | **Team Size:** 3 developers (Full-stack + QA) | **Fibonacci Scale:** 1 · 2 · 3 · 5 · 8

4.1 Sprint 1 — Foundation

Attribute	Detail
Sprint Goal	Establish secure access control and user administration capabilities
Duration	2 weeks
Team Capacity	12 Story Points
Stories	US-001 (3 SP) · US-002 (5 SP)
Total Committed	8 SP
Deliverables	Working login system · User management interface · RBAC · Session management

4.2 Sprint 2 — Patient Management

Attribute	Detail
Sprint Goal	Enable complete CRUD operations for patient records with search capability
Duration	2 weeks
Team Capacity	10 Story Points
Stories	US-003 (5 SP) · US-004 (3 SP) · US-005 (2 SP)

Attribute	Detail
Total Committed	10 SP
Deliverables	Child registration form · Paginated list with search · Edit/delete with validation · Duplicate detection

4.3 Sprint 3 — Visit Scheduling

Attribute	Detail
Sprint Goal	Implement complete visit scheduling with conflict detection and daily views
Duration	2 weeks
Team Capacity	10 Story Points
Stories	US-007 (5 SP) · US-009 (5 SP)
Total Committed	10 SP
Deliverables	Visit scheduling with time conflict detection · Today's visit dashboard · Visit history with filters

4.4 Sprint 4 — Consultation Management

Attribute	Detail
Sprint Goal	Link consultations to visits with full history tracking
Duration	2 weeks
Team Capacity	14 Story Points
Stories	US-011 (5 SP) · US-012 (3 SP) · US-006 (3 SP) · US-010 (3 SP)
Total Committed	14 SP

Attribute	Detail
Deliverables	Consultation form linked to visits · Expandable history view · Diagnosis recording

4.5 Sprint 5 — Prescription Management

Attribute	Detail
Sprint Goal	Enable complete prescription lifecycle with printing capability
Duration	2 weeks
Team Capacity	14 Story Points
Stories	US-013 (5 SP) · US-014 (3 SP) · US-015 (3 SP) · US-016 (3 SP)
Total Committed	14 SP
Deliverables	Prescription CRUD with role enforcement · Print-ready A4 template · Filterable prescription log

4.6 Sprint 6 — Vaccination Management

Attribute	Detail
Sprint Goal	Implement vaccination tracking with catalog management and overdue highlighting
Duration	2 weeks
Team Capacity	10 Story Points
Stories	US-017 (5 SP) · US-018 (5 SP)
Total Committed	10 SP
Deliverables	Vaccine catalog CRUD · Vaccination recording with batch tracking · History with overdue alerts

4.7 Sprint 7 — Financial Management

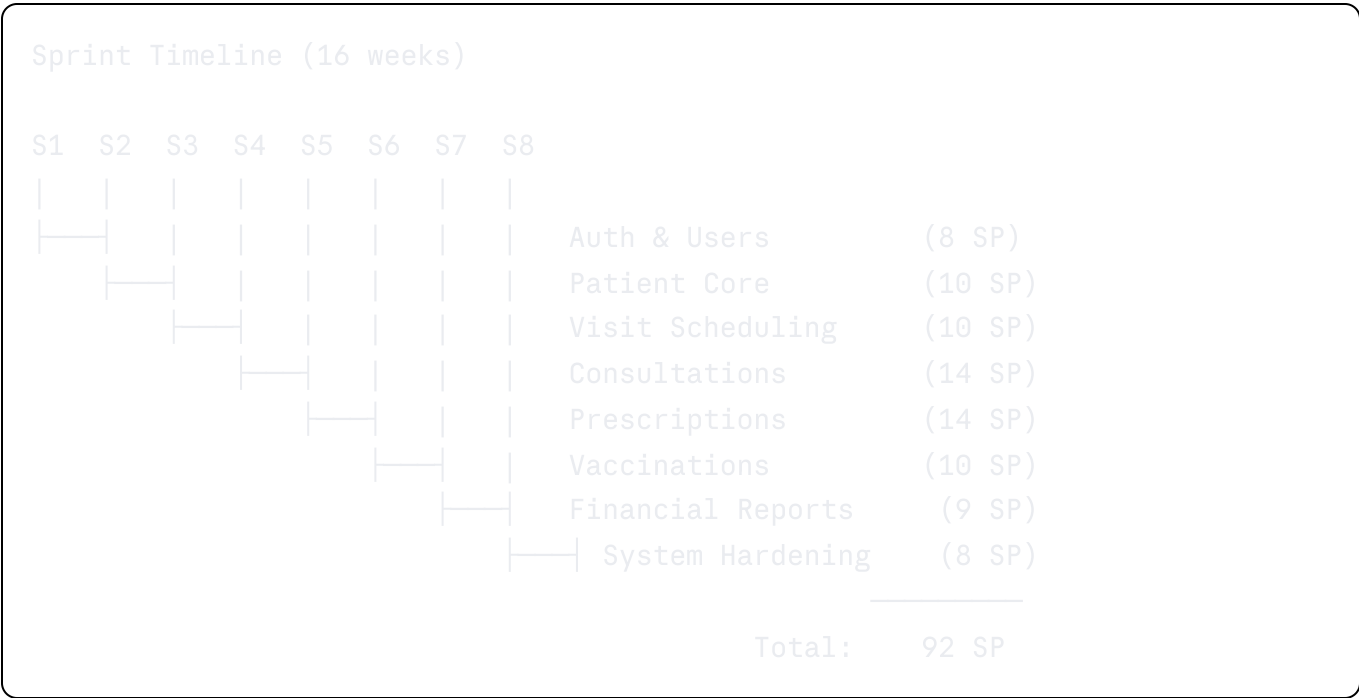
Attribute	Detail
Sprint Goal	Provide financial oversight with daily/monthly summaries and strict access control
Duration	2 weeks
Team Capacity	9 Story Points
Stories	US-021 (3 SP) · US-022 (3 SP) · US-023 (3 SP)
Total Committed	9 SP
Deliverables	Daily/monthly financial summaries with comparison · Role-based access enforcement

4.8 Sprint 8 — System Hardening

Attribute	Detail
Sprint Goal	Ensure production readiness with error handling, audit logging, and final polish
Duration	2 weeks
Team Capacity	8 Story Points
Stories	US-024 (3 SP) · US-025 (5 SP)
Total Committed	8 SP
Deliverables	Global error handling · Audit logging system · Final bug fixes · Deployment preparation

4.9 Project Velocity Summary

Metric	Planned	Notes
Total Story Points	92 SP	Corrected from v1.0 (was 71)
Sprint Count	8 sprints	2-week sprints
Average Velocity	11.5 SP / sprint	Stable throughput
Project Duration	16 weeks	~4 months calendar time
Team Size	3 developers	Full-stack + QA



5. Definition of Done

| A Product Backlog Item is **DONE** only when **ALL** of the following criteria are met.

#	Criterion	Verified By
1	Code is written and follows project coding standards	Code review
2	Unit tests written with minimum 80% code coverage	Coverage report
3	Integration tests pass for all affected modules	Integration test run
4	All Acceptance Criteria are met and verified	Test execution log
5	Code review completed with no critical issues	Review tool
6	Functional testing completed on target browsers (Chrome · Firefox · Edge)	Test report
7	Documentation updated (user guide, technical docs)	Doc review
8	No known critical or high-severity defects	Defect tracker
9	Performance criteria met (page load < 3 seconds)	Performance test
10	Feature demonstrated in Sprint Review and accepted by Product Owner	Sprint Review minutes